

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 34 — Nitrogen Compounds

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Amines & Basicity · Phenylamine & Azo Dye Synthesis · Amides & Hydrolysis · Amino Acids, Zwitterions & Electrophoresis
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED**34.1 Primary & Secondary Amines: Basicity Trends**

— Comparison of basic strengths of ethylamine, ammonia, and phenylamine; inductive electron donation

34.2 Phenylamine & Azo Dye Coupling Reactions

— Preparation of phenylamine via reduction of nitrobenzene with Sn and concentrated HCl; electrophilic

34.3 Amides: Structure, Basicity & Hydrolysis

— Preparation from acyl chlorides; neutral character explained by resonance delocalisation of the n

34.4 Amino Acids, Zwitterions & Electrophoresis

— Dipolar zwitterion structure; amphoteric acid-base buffer properties; isoelectric point; separation

High-Frequency Core Repeats (Q41–Q50)

— The 10 most frequently tested Cambridge Paper 4 questions on Nitrogen Compounds from the past

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Relative Base Strengths of Ethylamine, Ammonia, and Phenylamine — 9701/42/M/J/23/Q10 [6 Marks]

HARD · Syllabus 34.1

The relative base strengths of three nitrogen-containing bases are shown in Fig. 1.1.

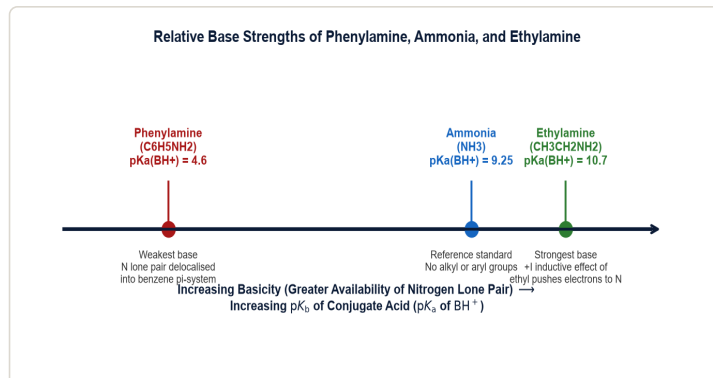


Fig. 1.1: Relative basicity and conjugate acid pK_a values for phenylamine, ammonia, and ethylamine.

- (a) Rank phenylamine, ammonia, and ethylamine in order of increasing base strength. [1]

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- (b) Explain why ethylamine is a stronger base than ammonia. [2]

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- (c) Explain in detail why phenylamine is a much weaker base than ammonia. [3]

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2 Diazotisation and Coupling Mechanism for Azo Dye Synthesis — 9701/41/O/N/22/Q8 [6 Marks]

HARD · Syllabus 34.2

The formation of 4-hydroxyazobenzene by electrophilic coupling is illustrated in Fig. 2.1.

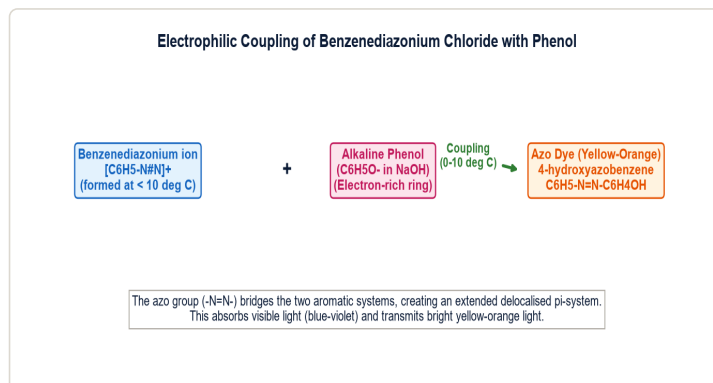


Fig. 2.1: Electrophilic coupling of benzenediazonium ion with phenoxide.

- (a) State the reagents and conditions needed to convert phenylamine into benzenediazonium chloride. [2]

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- (b) Explain why the diazonium solution must be kept below 10 °C, stating what happens if warmed. [2]
- (c) Explain why phenol is dissolved in aqueous sodium hydroxide before the coupling reaction, and draw the structure of the azo dye formed. [2]

3 Amino Acid Zwitterions and Gel Electrophoresis Separation — 9701/42/M/J/22/Q9 [6 Marks]

HARD · Syllabus 34.4

The behaviour of amino acids in aqueous solutions at different pH values is summarised in Fig. 3.1.

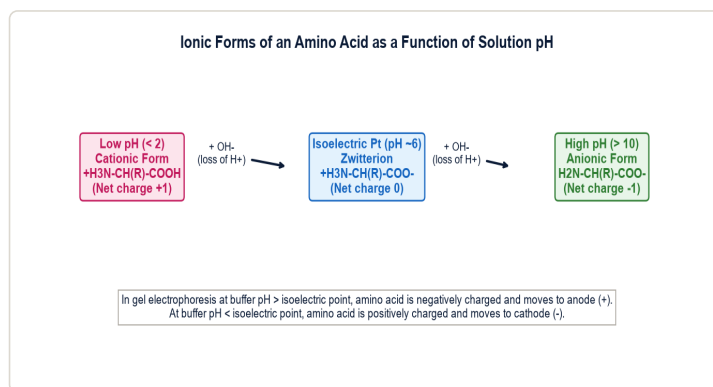


Fig. 3.1: Net electrical charge on an amino acid as a function of solution pH.

- (a) Draw the structural formula of alanine (2-aminopropanoic acid) in its zwitterionic form. [1]
- (b) Draw the structure of alanine at pH 1 and at pH 12. [2]
- (c) A mixture of alanine (isoelectric point 6.0), lysine (isoelectric point 9.7), and aspartic acid (isoelectric point 2.8) is separated by gel electrophoresis at a buffer pH of 6.0. Predict the direction of movement of each amino acid and explain your reasoning. [3]

4 Synthesis and Reactions of Phenylamine from Nitrobenzene — 9701/41/M/J/21/Q10 [6 Marks]

HARD · Syllabus 34.2

- (a) Give the reagents and conditions for the laboratory reduction of nitrobenzene to phenylamine, and explain why sodium hydroxide is added in the final step. [3]

- (b) State the observations when bromine water is added to aqueous phenylamine at room temperature. [2]
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- (c) Give the systematic IUPAC name and structure of the organic product formed in (b). [1]
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5 Acid and Alkaline Hydrolysis of Amides — 9701/42/O/N/20/Q8 [6 Marks]

HARD · Syllabus 34.3

- (a) Write the balanced chemical equation for the hydrolysis of ethanamide with hot dilute hydrochloric acid, naming both products. [3]
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- (b) Write the balanced chemical equation for the hydrolysis of ethanamide with hot aqueous sodium hydroxide, naming both products. [3]
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6 Synthesis of Primary Amines: Halogenoalkanes vs Reduction of Nitriles — 9701/43/M/J/23/Q9 [6 Marks]

HARD · Syllabus 34.1

- (a) Describe how bromoethane can be converted to ethylamine in a single step, stating reagents and essential conditions. [2]
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- (b) Explain why the reaction in (a) produces a mixture of primary, secondary, and tertiary amines, and state how the yield of primary amine can be maximised. [2]
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- (c) Describe a two-step synthesis to convert bromoethane into propylamine, stating all reagents and conditions. [2]
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7 Peptide Linkages, Polypeptides, and Hydrolysis of Proteins — 9701/42/F/M/22/Q10 [6 Marks]

HARD · Syllabus 34.4

A dipeptide is formed from glycine (2-aminoethanoic acid) and valine (2-amino-3-methylbutanoic acid).

- (a) Draw the structural formulas of the two isomeric dipeptides that can be formed from glycine and valine. [2]
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- (b) Identify the peptide linkage in one of the dipeptides and explain why it is planar. [2]

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- (c) State the reagents and conditions used to hydrolyse proteins completely into their constituent amino acids in the laboratory. [2]

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8 Electrophoresis: Influence of Buffer pH on Separation of Peptides — 9701/41/O/N/23/Q9 [6 Marks]

HARD · Syllabus 34.4

- (a) Define the term isoelectric point of an amino acid. [1]

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- (b) Glutamic acid has pKa values of 2.19 (–COOH), 4.25 (side-chain –COOH), and 9.67 (–NH₃⁺). Deduce its overall charge at pH 1.0, pH 7.0, and pH 12.0. [3]

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- (c) Explain how the voltage and duration of the electric current affect the separation of spots in gel electrophoresis. [2]

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9 Multi-Step Synthesis: Phenylamine to Paracetamol (4-Acetamidophenol) — 9701/42/M/J/20/Q8 [6 Marks]

HARD · Syllabus 34.3

- (a) Paracetamol is prepared from 4-aminophenol. Write the equation for the acylation of 4-aminophenol with ethanoyl chloride. [2]

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- (b) Explain why the amino group (–NH₂) is acylated preferentially rather than the phenolic hydroxy group (–OH). [2]

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- (c) State the reagents and conditions to prepare 4-aminophenol starting from nitrobenzene. [2]

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10 Basicity of Primary, Secondary, and Tertiary Aliphatic Amines — 9701/41/M/J/19/Q9 [6 Marks]

HARD · Syllabus 34.1

- (a) In the gas phase, the base strength of methylamines increases in the order: $\text{NH}_3 < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH} < (\text{CH}_3)_3\text{N}$. Explain this trend. [2]
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- (b) In aqueous solution, dimethylamine is a stronger base than trimethylamine. Explain why the basicity of trimethylamine decreases in water. [3]
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- (c) Write the equilibrium equation and expression for the base dissociation constant, K_b , of ethylamine in water. [1]
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11 Quantitative Analysis: Acid-Base Back Titration of an Insoluble Amine — 9701/42/O/N/21/Q9 [6 Marks]

HARD · Syllabus 34.1

A 1.395 g sample of an impure aromatic amine, $\text{C}_6\text{H}_5\text{NH}_2$ ($M_r = 93.0$), is dissolved in 50.00 cm^3 of $0.500 \text{ mol dm}^{-3}$ standard $\text{HCl}(\text{aq})$. The amine reacts completely: $\text{C}_6\text{H}_5\text{NH}_2 + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$. The excess unreacted HCl requires 28.40 cm^3 of $0.400 \text{ mol dm}^{-3}$ NaOH for complete neutralisation.

- (a) Calculate the initial moles of HCl added to the amine sample. [1]
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- (b) Calculate the moles of NaOH required to neutralise the unreacted HCl , and hence find the moles of HCl that reacted with the amine. [2]
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- (c) Calculate the mass of pure phenylamine in the sample and its percentage purity. [3]
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12 Comparison of Phenylamine, N-Methylphenylamine, and Diphenylamine — 9701/42/M/J/18/Q9 [6 Marks]

HARD · Syllabus 34.1

- (a) Draw the skeletal formula of diphenylamine, $(\text{C}_6\text{H}_5)_2\text{NH}$. [1]
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- (b) Explain why diphenylamine is an exceptionally weak base, even weaker than phenylamine. [3]

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- (c) Predict and explain whether N-methylphenylamine is a stronger or weaker base than phenylamine. [2]

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13 Optical Activity in Amino Acids and Separation of Enantiomers — 9701/41/O/N/18/Q9 [6 Marks]

HARD · Syllabus 34.4

- (a) State which 2-amino acid does not exhibit optical isomerism, giving a reason based on structure. [2]

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- (b) Draw the 3D stereochemical formulas of the two enantiomers of alanine, showing their non-superimposable mirror image relationship. [2]

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- (c) Explain why synthetic amino acids produced in a laboratory are optically inactive, whereas naturally occurring amino acids are optically active. [2]

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14 Synthesis of Nylon 6,6 and Comparison with Kevlar — 9701/42/F/M/20/Q9 [6 Marks]

HARD · Syllabus 34.3

- (a) State the systematic IUPAC names of the two monomers used to manufacture Nylon 6,6. [2]

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- (b) Draw the repeat unit of Kevlar, which is formed from benzene-1,4-diamine and benzene-1,4-dicarboxylic acid. [2]

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- (c) Explain why Kevlar has an exceptionally high tensile strength and thermal stability compared to Nylon 6,6. [2]

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15 Coupling of Diazonium Salts with Aromatic Amines — 9701/41/M/J/17/Q8 [6 Marks]

HARD · Syllabus 34.2

- (a) Benzenediazonium chloride can couple with N,N-dimethylphenylamine. Write the structural formula of the azo compound formed. [2]

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- (b) State the pH conditions required for coupling with amines compared to coupling with phenol. [2]

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- (c) Explain why coupling occurs at the 4-position (para) of N,N-dimethylphenylamine rather than the 2-position (ortho). [2]

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16 Solubility and Melting Points of Amino Acids vs Hydroxy Acids — 9701/42/O/N/17/Q9 [6 Marks]

HARD · Syllabus 34.4

Compare the physical properties of aminoethanoic acid (glycine, $M_r = 75.0$, melting point = $233\text{ }^{\circ}\text{C}$) and 2-hydroxyethanoic acid (glycolic acid, $M_r = 76.0$, melting point = $75\text{ }^{\circ}\text{C}$).

- (a) Explain the enormous difference in melting points between glycine and glycolic acid. [3]

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- (b) Explain why glycine is soluble in water but insoluble in non-polar organic solvents such as hexane. [3]

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Reaction of Ethylamine with Dilute Acids and Complex Ion Formation — 9701/42/M/J/23/Q10(b) [4 Marks] MEDIUM · Syllabus 34.1

- (a) Write the balanced chemical equation for the reaction of ethylamine with dilute sulfuric acid. [2]

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- (b) When ethylamine is added dropwise to aqueous copper(II) sulfate, a pale blue precipitate forms initially which dissolves in excess ethylamine to form a deep blue solution. Explain these observations. [2]

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18 Comparison of Reactivity of Phenylamine vs Benzene towards Bromine — 9701/41/O/N/22/Q8(b) [4 Marks] MEDIUM · Syllabus 34.2

- (a) State why benzene does not react with bromine water at room temperature. [2]

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- (b) Explain why phenylamine reacts vigorously with bromine water at room temperature without a catalyst. [2]
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19 Buffering Action of Zwitterionic Amino Acids — 9701/42/M/J/22/Q9(b) [4 Marks] MEDIUM · Syllabus 34.4

- (a) Write an equation showing how the zwitterion of glycine resists a decrease in pH when H^+ ions are added. [2]
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- (b) Write an equation showing how the zwitterion resists an increase in pH when OH^- ions are added. [2]
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20 Comparison of Phenylamine and Ethylamine with Ethanoyl Chloride — 9701/41/M/J/21/Q10(b) [4 Marks] MEDIUM · Syllabus 34.3

- (a) Draw the structural formula of the product formed when phenylamine reacts with ethanoyl chloride. [2]
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- (b) Name this product and state the type of bond formed between carbon and nitrogen. [2]
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21 Alkaline Hydrolysis of Amides: Identification of Volatile Product — 9701/42/O/N/20/Q8(b) [4 Marks] MEDIUM · Syllabus 34.3

- (a) When an unknown compound X is boiled with aqueous sodium hydroxide, an alkaline gas is evolved that turns damp red litmus paper blue. Identify the functional group present in X. [1]
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- (b) If compound X has molecular formula C_3H_7NO , deduce the two possible structural formulas for X. [2]
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- (c) State how the two isomers in (b) could be distinguished using infrared spectroscopy. [1]
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22 Synthesis of Quaternary Ammonium Salts and Surfactant Action — 9701/43/M/J/23/Q9(b) [4 Marks] MEDIUM · Syllabus 34.1

- (a) Write the chemical equation for the formation of tetraethylammonium bromide from triethylamine and bromoethane. [2]
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- (b) Explain why quaternary ammonium compounds with long hydrocarbon chains are used as cationic surfactants and fabric softeners. [2]
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23 Isoelectric Points and Structure of Lysine at Various pH Values — 9701/42/F/M/22/Q10(b) [4 Marks] MEDIUM · Syllabus 34.4

- (a) Lysine has two amino groups and one carboxyl group. Explain why its isoelectric point (9.7) is in the basic range. [2]
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- (b) Draw the structural formula of lysine at pH 7.0, indicating the charges on all functional groups. [2]
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24 Action of Nitrous Acid on Primary Aliphatic vs Aromatic Amines — 9701/41/O/N/23/Q9(b) [4 Marks] MEDIUM · Syllabus 34.2

- (a) Describe the visible observation and identify the gas evolved when ethylamine is treated with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) at 5°C . [2]
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- (b) Contrast this with the behaviour of phenylamine under the exact same conditions. [2]
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25 Amide Resonance and Lack of Basicity — 9701/42/M/J/20/Q8(b) [4 Marks] MEDIUM · Syllabus 34.3

- (a) Draw two resonance structures for ethanamide showing the delocalisation of the nitrogen lone pair. [2]
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- (b) Explain why ethanamide does not form a salt when mixed with dilute hydrochloric acid at room temperature. [2]
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26 Reduction of Nitriles and Amides: Comparing Synthetic Routes — 9701/41/M/J/19/Q9(b) [4 Marks] MEDIUM · Syllabus 34.1

- (a) Write an equation for the reduction of ethanenitrile to ethylamine using $[\text{H}]$ to represent the reducing agent. [1]
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- (b) Write an equation for the reduction of ethanamide to ethylamine using $[H]$. [1]
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- (c) Identify a suitable reducing agent for both reactions and state the required solvent. [2]
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27 Separation of Amino Acids by Two-Dimensional Thin-Layer Chromatography — 9701/42/O/N/21/Q9(b) [4 Marks] Syllabus 34.4

- (a) Explain why two-dimensional chromatography is necessary to separate complex mixtures of amino acids. [2]
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- (b) Name the locating agent used to visualise colourless amino acid spots on a chromatogram and state the colour observed. [2]
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28 Bromination of Phenylamine: Stoichiometry and Observations — 9701/42/M/J/18/Q9(b) [4 Marks] MEDIUM · Syllabus 34.2

- (a) Write the balanced chemical equation for the reaction of phenylamine with aqueous bromine. [2]
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- (b) State two visible observations that confirm this reaction has occurred. [2]
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29 Peptide Bond Hydrolysis: Acid-Catalysed vs Alkaline — 9701/41/O/N/18/Q9(b) [4 Marks] MEDIUM · Syllabus 34.4

- (a) Draw the structural formula of the product formed from the alanyl residue when glycylalanine is hydrolysed by boiling 6 mol dm⁻³ HCl. [2]
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- (b) Draw the structural formula of the product formed from the alanyl residue when glycylalanine is hydrolysed by boiling aqueous NaOH. [2]
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30 Polyamide Degradability: Chemical Hydrolysis of Nylon — 9701/42/F/M/20/Q9(b) [4 Marks] MEDIUM · Syllabus 34.3

- (a) Explain why clothing made of nylon weakens significantly when exposed to battery acid (sulfuric acid). [2]
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- (b) Write an equation showing the cleavage of one amide linkage in Nylon 6,6 by aqueous acid. [2]
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31 Chromophore in Methyl Orange Azo Indicator — 9701/41/M/J/17/Q8(b) [4 Marks]

MEDIUM · Syllabus 34.2

- (a) Identify the chromophore group common to all azo dyes. [1]
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- (b) Explain how the colour of methyl orange changes between acidic and alkaline conditions in terms of electronic delocalisation. [3]
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32 Reaction of Amino Acids with Nitrous Acid: Nitrogen Gas Evolution — 9701/42/O/N/17/Q9(b) [4 Marks]

MEDIUM · Syllabus 34.4

- (a) When glycine is reacted with nitrous acid ($\text{NaNO}_2 + \text{HCl}$), effervescence is observed and a hydroxy acid is produced. Write the balanced equation. [2]
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- (b) Explain how measuring the volume of nitrogen gas evolved in this reaction can be used for quantitative analysis of amino acids (Van Slyke method). [2]
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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Classification of Amines: Primary, Secondary, Tertiary — 9701/42/M/J/23/Q10(a) [2 Marks]

EASY · Syllabus 34.1

- (a) Classify diethylamine and triethylamine as primary, secondary, or tertiary amines. [2]
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34 Formula of Benzenediazonium Cation — 9701/41/O/N/22/Q8(a) [2 Marks]

EASY · Syllabus 34.2

- (a) Write the formula of the benzenediazonium cation, displaying the bonding between the two nitrogen atoms. [2]
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35 Definition of Zwitterion — 9701/42/M/J/22/Q9(a) [2 Marks]

EASY · Syllabus 34.4

- (a) Explain what is meant by the term zwitterion, giving one structural example. [2]
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36 Preparation of Primary Amides — 9701/41/M/J/21/Q10(a) [2 Marks]

EASY · Syllabus 34.3

- (a) State the organic reactant and inorganic reactant used to prepare propanamide in a single step at room temperature. [2]

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37 Acid-Base Nature of Ethanamide — 9701/42/O/N/20/Q8(a) [2 Marks]

EASY · Syllabus 34.3

- (a) State whether ethanamide is acidic, basic, or neutral in aqueous solution, giving a brief reason. [2]

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38 Conditions for Diazotisation — 9701/43/M/J/23/Q9(a) [2 Marks]

EASY · Syllabus 34.2

- (a) State the temperature range required for the diazotisation of phenylamine and name the inorganic acid used. [2]

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39 Peptide Bond Displayed Formula — 9701/42/F/M/22/Q10(a) [2 Marks]

EASY · Syllabus 34.4

- (a) Draw the displayed formula of a peptide linkage, showing all bonds. [2]

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40 Direction of Migration in Electrophoresis — 9701/41/O/N/23/Q9(a) [2 Marks]

EASY · Syllabus 34.4

- (a) In gel electrophoresis at pH 12, an amino acid exists as $\text{H}_2\text{N}-\text{CHR}-\text{COO}^-$. State which electrode it migrates towards and explain why. [2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 Core Repeat 1: Basicity Hierarchy of Ammonia, Alkylamines, and Phenylamine — 9701/42/M/J/22/Q8 [6 Marks]

EASY · Syllabus 34.1

The relative basicity of nitrogen bases is among the most regularly repeated structured questions in Cambridge Paper 4.

- (a) State the definition of a Bronsted-Lowry base and write the equation for the reaction of ethylamine with water. [2]

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- (b) Explain why ethylamine is a stronger base than ammonia, referring to inductive effects. [2]

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- (c) Explain why phenylamine is a substantially weaker base than ammonia, referring to orbital overlap. [2]

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42 Core Repeat 2: Full Synthesis and Mechanism of Azo Dye Coupling — 9701/41/O/N/21/Q9 [6 Marks]

HARD · Syllabus 34.2

- (a) Describe how phenylamine is converted to benzenediazonium chloride, specifying all reagents, equations, and temperature controls. [3]

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- (b) Describe how the benzenediazonium ion reacts with an alkaline solution of phenol, stating the observation and giving the structural formula of the azo dye. [3]

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43 Core Repeat 3: Gel Electrophoresis Separation of Three Amino Acids at Two Buffer pH Values — 9701/42/M/J/21/Q10 [6 Marks]

The isoelectric points (pI) of three amino acids are:

- Serine: pI = 5.7
- Arginine: pI = 10.8
- Aspartic acid: pI = 2.8

- (a) Predict the direction of migration (towards anode +, cathode -, or remaining near origin) for each amino acid when electrophoresis is conducted at a buffer pH of 5.7. [3]

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- (b) Predict the direction of migration for each amino acid when electrophoresis is conducted at a buffer pH of 1.5. [3]

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44 Core Repeat 4: Multi-Step Synthetic Route to an Aromatic Amide from Benzene — 9701/42/O/N/19/Q9 [6 Marks]

HARD · Syllabus 34.3

Devise a four-step synthetic route to convert benzene, C_6H_6 , into N-phenylethanamide, $C_6H_5NHCOCH_3$.

- (a) Step 1 converts benzene into nitrobenzene. State the reagents and temperature. [2]

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- (b) Step 2 reduces nitrobenzene to phenylamine. State the reagents, conditions, and the work-up reagent. [2]
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- (c) Step 3 reacts phenylamine with an acylating agent to form N-phenylethanamide. Name this agent, write the equation, and state the byproduct. [2]
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45 Core Repeat 5: Acid and Alkaline Hydrolysis of a Dipeptide — 9701/42/M/J/23/Q10(c) [4 Marks]

MEDIUM · Syllabus 34.4

Glycylalanine has the formula $\text{H}_2\text{N}-\text{CH}_2-\text{CO}-\text{NH}-\text{CH}(\text{CH}_3)-\text{COOH}$.

- (a) Draw the structures of the organic products formed when glycylalanine is boiled with excess 6 mol dm^{-3} hydrochloric acid. [2]
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- (b) Draw the structures of the organic products formed when glycylalanine is boiled with excess aqueous sodium hydroxide. [2]
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46 Core Repeat 6: Synthesis and Basicity of Secondary and Tertiary Amines — 9701/41/O/N/22/Q8(c) [4 Marks]

MEDIUM · Syllabus 34.1

- (a) Write the balanced chemical equation for the reaction of ethylamine with bromoethane to form diethylamine. [2]
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- (b) Explain why diethylamine can react further with bromoethane and state the final product formed if bromoethane is in large excess. [2]
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47 Core Repeat 7: Azo Coupling with 2-Naphthol — 9701/42/F/M/21/Q10(b) [4 Marks]

MEDIUM · Syllabus 34.2

- (a) State the observations when benzenediazonium chloride is added to an ice-cold alkaline solution of 2-naphthol. [2]
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- (b) Explain why azo dyes are intensely coloured and resistant to fading in sunlight. [2]
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48 Core Repeat 8: Distinction Between Phenylamine, Benzamide, and Benzyl Chloride — 9701/41/M/J/20/Q9(a) [4 Marks] EASY · Syllabus 34.3

Three unlabelled bottles contain phenylamine ($\text{C}_6\text{H}_5\text{NH}_2$), benzamide ($\text{C}_6\text{H}_5\text{CONH}_2$), and benzyl chloride ($\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$).

- (a) Describe a chemical test that will identify phenylamine immediately. [2]

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- (b) Describe a chemical test that will identify benzamide. [2]

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49 Core Repeat 9: Observation for Phenylamine Bromination — 9701/42/M/J/23/Q10(e) [2 Marks] EASY · Syllabus 34.2

- (a) State two visible observations when excess aqueous bromine is added to phenylamine. [2]

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50 Core Repeat 10: Isoelectric Point and Net Charge — 9701/41/O/N/23/Q9(e) [2 Marks] EASY · Syllabus 34.4

- (a) State the net charge on an amino acid when the solution pH is strictly less than its isoelectric point, and deduce which electrode it migrates to during electrophoresis. [2]

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Mark Scheme — Topic 34 — Nitrogen Compounds

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Phenylamine < Ammonia < Ethylamine [1].	1
1(b)	The ethyl group exerts an electron-donating (+I) inductive effect towards the nitrogen atom [1]; This increases the electron density of the lone pair on nitrogen, making it more available to accept a proton (H ⁺) [1].	2
1(c)	In phenylamine, the p-orbital lone pair of electrons on the nitrogen atom overlaps with the delocalised π -electron cloud of the benzene ring [1]; The lone pair is delocalised into the aromatic ring [1]; This significantly decreases electron density on nitrogen, making the lone pair far less available to coordinate with a proton [1].	3
2(a)	Sodium nitrite (NaNO ₂) and hydrochloric acid (HCl) (or nitrous acid, HNO ₂) [1]; Temperature between 0 °C and 10 °C [1].	2
2(b)	The benzenediazonium ion is thermally unstable above 10 °C [1]; It decomposes / hydrolyses to form phenol and nitrogen gas (N ₂) [1].	2
2(c)	NaOH converts phenol into the phenoxide ion (C ₆ H ₅ O ⁻), which carries a full negative charge and is far more activated / nucleophilic towards the diazonium electrophile [1]; Structure of 4-hydroxyazobenzene showing C ₆ H ₅ -N=N-C ₆ H ₄ -OH [1].	2
3(a)	+H ₃ N-CH(CH ₃)-COO ⁻ [1].	1
3(b)	At pH 1: +H ₃ N-CH(CH ₃)-COOH (cationic form) [1]; At pH 12: H ₂ N-CH(CH ₃)-COO ⁻ (anionic form) [1].	2
3(c)	Alanine has net charge zero at pH 6.0 and remains at the origin / does not migrate [1]; Lysine (pI 9.7) is below its isoelectric point, carries a net positive charge (+1), and migrates towards the cathode (-) [1]; Aspartic acid (pI 2.8) is above its isoelectric point, carries a net negative charge (-1), and migrates towards the anode (+) [1].	3
4(a)	Tin (Sn) and concentrated hydrochloric acid (conc. HCl) [1]; Heated under reflux [1]; NaOH is added to deprotonate the phenylammonium salt (C ₆ H ₅ NH ₃ ⁺ Cl ⁻) to liberate free phenylamine [1].	3
4(b)	Orange/brown bromine water is decolourised [1]; A white precipitate is formed [1].	2
4(c)	2,4,6-tribromophenylamine (or 2,4,6-tribromoaniline) with correct structure [1].	1
5(a)	CH ₃ CONH ₂ + H ₂ O + HCl → CH ₃ COOH + NH ₄ Cl (or + H ⁺ + H ₂ O → CH ₃ COOH + NH ₄ ⁺) [2]; Ethanoic acid and ammonium chloride [1].	3
5(b)	CH ₃ CONH ₂ + NaOH → CH ₃ COONa + NH ₃ (or + OH ⁻ → CH ₃ COO ⁻ + NH ₃) [2]; Sodium ethanoate and ammonia [1].	3
6(a)	Excess concentrated ammonia (NH ₃) in ethanol [1]; Heated in a sealed tube under pressure [1].	2
6(b)	The ethylamine formed retains a lone pair on nitrogen and acts as a nucleophile to attack unreacted bromoethane, undergoing further substitution [1]; Use a large excess of ammonia to ensure ammonia outcompetes ethylamine for bromoethane [1].	2
6(c)	Step 1: Potassium cyanide (KCN) in ethanol, heated under reflux to form propanenitrile [1]; Step 2: Lithium tetrahydridoaluminate (LiAlH ₄) in dry ether (or H ₂ with Ni catalyst) to reduce propanenitrile to propylamine [1].	2
7(a)	Gly-Val: H ₂ N-CH ₂ -CO-NH-CH(CH ₃) ₂ -COOH [1]; Val-Gly: H ₂ N-CH(CH ₃) ₂ -CO-NH-CH ₂ -COOH [1].	2
7(b)	-CO-NH- group highlighted [1]; The nitrogen lone pair delocalises into the carbonyl π -bond, giving the C-N bond partial double bond character which prevents free rotation and holds the atoms in a planar geometry [1].	2
7(c)	Concentrated hydrochloric acid (6 mol dm ⁻³ HCl) [1]; Heated under reflux for 24 hours at 110 °C [1].	2
8(a)	The specific pH at which an amino acid or peptide exists predominantly as a zwitterion with zero net electrical charge [1].	1
8(b)	At pH 1.0: net charge is +1 (+H ₃ N-CH(COOH)-CH ₂ CH ₂ COOH) [1]; At pH 7.0: net charge is -1 (+H ₃ N-CH(COO ⁻)-CH ₂ CH ₂ COO ⁻) [1]; At pH 12.0: net charge is -2 (H ₂ N-CH(COO ⁻)-CH ₂ CH ₂ COO ⁻) [1].	3
8(c)	Higher voltage increases the electrostatic force and migration speed, but excessive voltage causes Joule heating that distorts bands [1]; Longer duration allows greater distance of migration and better resolution between species of similar charge-to-mass ratio, but excessive time leads to band broadening by diffusion [1].	2
9(a)	4-HOC ₆ H ₄ NH ₂ + CH ₃ COCl → 4-HOC ₆ H ₄ NHCOCH ₃ + HCl [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
9(b)	Nitrogen is less electronegative than oxygen and holds its lone pair less tightly [1]; Therefore, the -NH ₂ group is a significantly more powerful nucleophile than the -OH group and attacks the carbonyl carbon much more rapidly [1].	2
9(c)	Mononitration of phenol (or hydroxylation of nitrobenzene) followed by reduction with Sn and concentrated HCl [2].	2
10(a)	Each additional methyl group exerts an electron-donating (+I) inductive effect [1]; This progressively increases electron density on the nitrogen lone pair, making it increasingly able to coordinate a proton in the absence of solvent [1].	2
10(b)	In aqueous solution, the stability of the conjugate cation depends on hydration via hydrogen bonding with water molecules [1]; (CH ₃) ₃ NH ⁺ has only one hydrogen atom attached to nitrogen to form hydrogen bonds, compared to two in (CH ₃) ₂ NH ₂ ⁺ [1]; In addition, three bulky methyl groups create steric hindrance that obstructs the approach of hydrating water molecules, destabilising the cation [1].	3
10(c)	CH ₃ CH ₂ NH ₂ + H ₂ O ⇌ CH ₃ CH ₂ NH ₃ ⁺ + OH ⁻ ; K _b = [CH ₃ CH ₂ NH ₃ ⁺][OH ⁻] / [CH ₃ CH ₂ NH ₂] [1].	1
11(a)	Moles of HCl initial = 0.05000 x 0.500 = 2.500 x 10 ⁻² mol [1].	1
11(b)	Moles of NaOH = 0.02840 x 0.400 = 1.136 x 10 ⁻² mol [1]; Moles of HCl reacted = 2.500 x 10 ⁻² - 1.136 x 10 ⁻² = 1.364 x 10 ⁻² mol [1].	2
11(c)	Moles of C ₆ H ₅ NH ₂ = 1.364 x 10 ⁻² mol [1]; Mass of C ₆ H ₅ NH ₂ = 1.364 x 10 ⁻² x 93.0 = 1.269 g [1]; Percentage purity = (1.269 / 1.395) x 100 = 90.9% [1].	3
12(a)	Two benzene rings attached to a central -NH- group [1].	1
12(b)	In diphenylamine, the nitrogen lone pair can overlap and delocalise into two separate benzene π-rings simultaneously [1]; This extensive double delocalisation drastically reduces electron density on the nitrogen atom [1]; Consequently, the lone pair has virtually no tendency to accept a proton [1].	3
12(c)	N-methylphenylamine is slightly stronger than phenylamine [1]; The electron-donating (+I) inductive effect of the methyl group partially offsets the delocalisation into the phenyl ring, increasing electron density on nitrogen [1].	2
13(a)	Glycine (2-aminoethanoic acid) [1]; The alpha-carbon is bonded to two identical hydrogen atoms, so it lacks a chiral / asymmetric carbon centre [1].	2
13(b)	Two 3D tetrahedral structures drawn with wedge and dash bonds as non-superimposable mirror images around the chiral carbon [2].	2
13(c)	Laboratory chemical synthesis proceeds via planar intermediates (e.g. carbonyls or carbocations) where attack by reagents is equally probable from either side, producing an equimolar racemic mixture [1]; In biological systems, enzymes have asymmetric chiral active sites that catalyse stereospecific reactions producing exclusively one optical isomer (usually L-amino acids) [1].	2
14(a)	Hexane-1,6-diamine (or 1,6-diaminohexane) [1]; Hexanedioic acid (or hexanedioyl dichloride) [1].	2
14(b)	-[NH-C ₆ H ₄ -NH-CO-C ₆ H ₄ -CO]- with correct 1,4-disubstituted aromatic rings and amide linkages [2].	2
14(c)	The rigid aromatic rings keep the polymer chains straight and planar, allowing tight parallel alignment and packing [1]; Extensive and regular intermolecular hydrogen bonding between aligned amide groups, combined with π-π stacking between aromatic rings, provides immense tensile strength [1].	2
15(a)	C ₆ H ₅ -N=N-C ₆ H ₄ -4-N(CH ₃) ₂ [2].	2
15(b)	Coupling with amines is carried out in weakly acidic to neutral pH (~4–7) [1]; Coupling with phenol requires alkaline pH (~9–11) to generate the phenoxide ion [1].	2
15(c)	The bulky -N(CH ₃) ₂ group creates significant steric hindrance around the 2- and 6-positions [1]; The 4-position is sterically unhindered and fully activated by the electron-donating nitrogen lone pair [1].	2
16(a)	Glycine exists in the solid state as a dipolar zwitterion (+H ₃ N-CH ₂ -COO ⁻) [1]; It forms an ionic lattice held together by very strong electrostatic attractions (ionic bonds) between oppositely charged ions, requiring massive energy to overcome [1]; Glycolic acid exists as neutral covalent molecules held together only by hydrogen bonding and London dispersion forces, which are much weaker than ionic bonds [1].	3

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
16(b)	The ionic zwitterions of glycine are strongly hydrated by ion-dipole interactions with polar water molecules, releasing hydration energy to overcome lattice energy [1]; Hexane molecules are non-polar and can only form weak London dispersion forces with the zwitterions [1]; These weak interactions cannot compensate for the large energy required to break the ionic lattice of glycine [1].	3
17(a)	$2\text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{SO}_4 \rightarrow (\text{CH}_3\text{CH}_2\text{NH}_3^+)_2\text{SO}_4^{2-}$ (or $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}^+ \rightarrow \text{CH}_3\text{CH}_2\text{NH}_3^+$) [2].	2
17(b)	Initially, ethylamine acts as a base generating OH^- ions to precipitate pale blue $\text{Cu}(\text{OH})_2(\text{s})$ [1]; In excess, ethylamine acts as a monodentate ligand, replacing water to form the deep blue complex $[\text{Cu}(\text{CH}_3\text{CH}_2\text{NH}_2)_4(\text{H}_2\text{O})_2]^{2+}$ [1].	2
18(a)	Benzene has a stable delocalised aromatic π -system and insufficient electron density to polarise non-polar Br_2 molecules [2].	2
18(b)	The lone pair on the nitrogen atom delocalises into the aromatic π -cloud, significantly activating the benzene ring [1]; The increased electron density is high enough to induce a dipole in Br_2 molecules without needing a halogen carrier [1].	2
19(a)	$+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^- + \text{H}^+ \rightarrow +\text{H}_3\text{N}-\text{CH}_2-\text{COOH}$ (the carboxylate group accepts H^+) [2].	2
19(b)	$+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^- + \text{OH}^- \rightarrow \text{H}_2\text{N}-\text{CH}_2-\text{COO}^- + \text{H}_2\text{O}$ (the ammonium group donates H^+) [2].	2
20(a)	$\text{CH}_3\text{CONH}-\text{C}_6\text{H}_5$ (N-phenylethanamide / acetanilide) [2].	2
20(b)	N-phenylethanamide [1]; Secondary amide bond (peptide link / $-\text{CONH}-$) [1].	1
21(a)	Amide group ($-\text{CONH}_2$ or $-\text{CONHR}$) [1].	1
21(b)	Propanamide, $\text{CH}_3\text{CH}_2\text{CONH}_2$ [1]; N-methylethanamide, $\text{CH}_3\text{CONHCH}_3$ [1].	2
21(c)	Primary amide (propanamide) displays two N-H stretching peaks at $3300\text{--}3500\text{ cm}^{-1}$, whereas secondary amide (N-methylethanamide) displays only a single N-H peak [1].	1
22(a)	$(\text{CH}_3\text{CH}_2)_3\text{N} + \text{CH}_3\text{CH}_2\text{Br} \rightarrow (\text{CH}_3\text{CH}_2)_4\text{N}^+ \text{Br}^-$ [2].	2
22(b)	They possess a long non-polar hydrophobic hydrocarbon tail that interacts with oils/fabrics and a positively charged hydrophilic head that interacts with water [1]; The positive charge binds strongly to negatively charged fabric/hair fibres, reducing static friction and giving softness [1].	2
23(a)	Lysine has a basic amino side-chain ($-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$) [1]; A high pH (excess OH^-) is needed to deprotonate the ammonium groups to reach the neutral zwitterionic form with net charge zero [1].	2
23(b)	$+\text{H}_3\text{N}-\text{CH}(\text{COO}^-)-(\text{CH}_2)_4-\text{NH}_3^+$ showing one $-\text{COO}^-$ and two $-\text{NH}_3^+$ groups (net charge +1) [2].	2
24(a)	Vigorous effervescence / bubbles of colourless gas [1]; Nitrogen gas, N_2 (aliphatic diazonium ion decomposes immediately to ethanol and N_2) [1].	2
24(b)	Phenylamine forms a stable clear solution of benzenediazonium chloride without effervescence [1]; The diazonium group is stabilised by delocalisation into the aromatic π -cloud at 5°C [1].	2
25(a)	Structure 1: $\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$ with lone pair on N [1]; Structure 2: $\text{CH}_3-\text{C}(\text{O}^-)=\text{N}+\text{H}_2$ with double bond to N and formal charges [1].	2
25(b)	The lone pair on nitrogen is delocalised into the π -orbital of the carbonyl group [1]; It is not sufficiently available to donate to a proton (H^+), so amides are neutral and do not protonate in dilute acid [1].	2
26(a)	$\text{CH}_3\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ [1].	1
26(b)	$\text{CH}_3\text{CONH}_2 + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O}$ [1].	1
26(c)	Lithium tetrahydridoaluminate, LiAlH_4 [1]; In dry ethoxyethane (dry ether) [1].	2
27(a)	Several amino acids may have very similar or identical R_f values in a single solvent system, leading to overlapping spots [1]; Rotating the plate by 90° and running in a second different solvent with different polarity separates the co-eluting amino acids [1].	2
27(b)	Ninhydrin spray [1]; Followed by gentle heating to develop purple / blue-violet spots [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
28(a)	$C_6H_5NH_2 + 3Br_2 \rightarrow C_6H_2Br_3NH_2 + 3HBr$ [2].	2
28(b)	The orange-brown colour of bromine is decolourised [1]; A white precipitate of 2,4,6-tribromophenylamine forms [1].	2
29(a)	$+H_3N-CH(CH_3)-COOH$ (alanine hydrochloride cation) [2].	2
29(b)	$H_2N-CH(CH_3)-COO^- Na^+$ (sodium alaninate anion) [2].	2
30(a)	Strong acid hydrolyses the amide ($-CO-NH-$) linkages along the polymer backbone, breaking the long chains into shorter fragments and destroying tensile strength [2].	2
30(b)	$-CO-NH- + H_2O + H^+ \rightarrow -COOH + -NH_3^+$ [2].	2
31(a)	The azo group / $-N=N-$ double bond [1].	1
31(b)	In acid, protonation occurs at one of the azo nitrogen atoms [1]; This modifies the extent of π -electron delocalisation and alters the energy gap E between HOMO and LUMO [1]; The molecule absorbs a different wavelength of visible light, shifting the observed complementary colour from yellow (alkali) to red (acid) [1].	3
32(a)	$H_2N-CH_2-COOH + HNO_2 \rightarrow HO-CH_2-COOH + N_2 + H_2O$ [2].	2
32(b)	1 mole of $-NH_2$ group produces exactly 1 mole of N_2 gas (1:1 stoichiometry) [1]; By measuring gas volume at known temperature and pressure, the moles of amino acid can be directly calculated via $pV = nRT$ [1].	2
33(a)	Diethylamine is a secondary amine [1]; Triethylamine is a tertiary amine [1].	2
34(a)	$[C_6H_5-N \equiv N]^+$ with a triple bond between the two nitrogen atoms and a positive charge on the diazonium group [2].	2
35(a)	An electrically neutral molecule carrying both a positive and a negative formal charge on different atoms simultaneously [1]; Example: $+H_3N-CH_2-COO^-$ [1].	2
36(a)	Propanoyl chloride, CH_3CH_2COCl [1]; Concentrated aqueous ammonia, NH_3 [1].	2
37(a)	Neutral [1]; The nitrogen lone pair is delocalised into the adjacent carbonyl group and is not available to accept a proton [1].	2
38(a)	0 °C to 10 °C [1]; Hydrochloric acid, HCl (or nitrous acid, HNO_2) [1].	2
39(a)	$-C(=O)-N(H)-$ showing the $C=O$ double bond, $C-N$ single bond, and $N-H$ single bond clearly [2].	2
40(a)	Migrates towards the anode (positive electrode) [1]; Because it carries a negative net electrical charge and is attracted to the opposite charge [1].	2
41(a)	A proton (H^+) acceptor [1]; $CH_3CH_2NH_2 + H_2O \rightleftharpoons CH_3CH_2NH_3^+ + OH^-$ [1].	2
41(b)	The ethyl group exerts an electron-donating (+I) inductive effect, increasing electron density on nitrogen and making the lone pair more available to accept H^+ [2].	2
41(c)	The p-orbital lone pair on nitrogen overlaps sideways with the delocalised π -electron system of the benzene ring [1]; The lone pair is delocalised into the ring, significantly reducing its availability to accept a proton [1].	2
42(a)	$NaNO_2$ and concentrated (or dilute) HCl (generating HNO_2) [1]; $C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+ Cl^- + 2H_2O$ [1]; Temperature kept between 0 °C and 10 °C to prevent decomposition [1].	3
42(b)	Alkaline solution of phenol (phenoxide ion, $C_6H_5O^-$) is cooled in ice and mixed with the diazonium solution [1]; An intense yellow-orange precipitate / colour is observed [1]; Structure of 4-hydroxyazobenzene: $C_6H_5-N=N-C_6H_4-OH$ [1].	3
43(a)	At pH 5.7: Serine is at its pI, has net charge 0, and remains at origin [1]; Arginine (pI 10.8) is below its pI, carries net positive charge, and migrates to cathode (-) [1]; Aspartic acid (pI 2.8) is above its pI, carries net negative charge, and migrates to anode (+) [1].	3
43(b)	At pH 1.5: All three amino acids are at a pH below their isoelectric points [1]; All three carry net positive charges ($+H_3N-CHR-COOH$) [1]; Therefore, all three migrate towards the cathode (-), with separation depending on their charge-to-mass ratios [1].	3
44(a)	Concentrated HNO_3 and concentrated H_2SO_4 at 55 °C [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
44(b)	Tin (Sn) and concentrated HCl, heated under reflux [1]; Followed by aqueous NaOH to liberate free phenylamine [1].	2
44(c)	Ethanoyl chloride, CH ₃ COCl (or ethanoic anhydride) [1]; C ₆ H ₅ NH ₂ + CH ₃ COCl → C ₆ H ₅ NHCOCH ₃ + HCl [1].	2
45(a)	+H ₃ N-CH ₂ -COOH and +H ₃ N-CH(CH ₃)-COOH (both amino groups protonated, carboxyls intact) [2].	2
45(b)	H ₂ N-CH ₂ -COO ⁻ and H ₂ N-CH(CH ₃)-COO ⁻ (both carboxyl groups deprotonated as sodium salts, amino groups neutral) [2].	2
46(a)	CH ₃ CH ₂ NH ₂ + CH ₃ CH ₂ Br → (CH ₃ CH ₂) ₂ NH + HBr (or + 2CH ₃ CH ₂ NH ₂ diethylamine + ethylammonium bromide) [2].	2
46(b)	Diethylamine has a lone pair on nitrogen and acts as a nucleophile to attack more bromoethane [1]; Final product: tetraethylammonium bromide, (CH ₃ CH ₂) ₄ N ⁺ Br ⁻ (quaternary ammonium salt) [1].	2
47(a)	A bright red-orange precipitate / dye (Sudan I) is formed immediately [2].	2
47(b)	The extended conjugated π-system spans both aromatic rings and the azo group, absorbing in the visible spectrum [1]; High delocalisation resonance energy imparts stability against photochemical degradation by UV light [1].	2
48(a)	Add bromine water [1]; Bromine water is decolourised and a white precipitate of 2,4,6-tribromophenylamine forms (neither benzamide nor benzyl chloride react) [1].	2
48(b)	Boil with aqueous sodium hydroxide [1]; An alkaline gas (ammonia, NH ₃) is evolved that turns damp red litmus paper blue [1].	2
49(a)	Orange/brown bromine water is decolourised [1]; A white precipitate is formed [1].	2
50(a)	Positive net charge (+1 or higher) [1]; Migrates towards the cathode (negative electrode) [1].	2